COMPOSITE MATERIAL DATASHEETS CFRP IM5/RTM6-NCF

MATERIAL IDENTIFICATION

<table>
<tr><td>Short informal name</td><td>IM5/RTM6-NCF</td></tr>
<tr><td>Commercial name</td><td>Hexcel HiMax IM5/V100 with RTM6</td></tr>
<tr><td>Specifications</td><td>This material is not qualified according Airbus standards and specifications. Design data proposed here shall be consider preliminary until the qualification and further tests programs planned are concluded. Most values inferred from similar materials and engineering judgements. Fiber Volume Fraction (FVF) considered is 58%.</td></tr>
<tr><td>Description</td><td>Carbon fiber multi-axial / non-crimp fabric (NCF) in epoxy resin. Produced by a liquid resin infusion (LRI) process. Service temperature: -55 to 90ºC. Cure: 120-135 minutes at 185 ± 5ºC. Available formats, with nominal 58% FVF, are:
<table>
<tr><th>N-axial</th><th>Orientations (º)</th><th>Thickness (mm)</th><th>FAW (g/m²)</th></tr>
<tr><td>UD</td><td>0</td><td>0.258</td><td>268</td></tr>
<tr><td>Bi-axial</td><td>0/90</td><td>0.516</td><td>536</td></tr>
<tr><td></td><td>90/0</td><td>0.516</td><td>536</td></tr>
<tr><td></td><td>+45/-45</td><td>0.516</td><td>536</td></tr>
<tr><td></td><td>-45/+45</td><td>0.516</td><td>536</td></tr>
</table></td></tr>
<tr><td>Version</td><td>0.1 (01/2025)</td></tr>
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REFERENCES

- [Airbus DS] Memorandum 2024 – 07.001 "CLEAN AVIATION – HERFUSE – NCF selection proposal based on available HEXCEL IM5 formats" (04/07/2024)
- [Airbus DS] Memorandum 2024 – 09.003 "CLEAN AVIATION – HERFUSE – NCF IM5 and RTM6 manufacturing information to feed Engineering Dossier for Partners" (03/10/2024)
- [Airbus DS] Stress Methods Manual, Vol. 1 (including HiTape/RTM6-UD material datasheet)
- [Hexcel] HextTow IM5 Carbon Fiber Product Data Sheet

PHYSICAL PROPERTIES

Cured Ply Thickness (*CPT*)	0.258 (*) mm
Density	1600 kg/m³
Tg-onset (wet)	TBD ºC
Max. Operational Limit (*MOL*)	TBD ºC

Coefficients of Thermal Expansion CTE (1/ºC)	Range	-55…20ºC	20…70ºC	70…120ºC
	α_1	0	0	0
	α_2	3.0E-5	3.0E-5	3.0E-5
	α_3	3.0E-5	3.0E-5	3.0E-5

(*) The nominal value of the CPT to be used in stress analysis is 0.258mm, corresponding to the basic uni-axial ply of the material in the different formats (including bi-axial packages of 2 plies). The rest of the design data shown also correspond to this basic UD ply. However, the laminates shall be built on stacking sequences consistent with the available uni-axial and multi-axial sub-laminate / formats listed in the *Description* field.

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12} -	K_B -
Values	150000	7000	3500	0.35	0.85

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13} -	ν_{23} -
Values	7000	3500	2000	0.35	0.5

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1, E_2 and G_{12} in buckling stability analyses
- Apparent modulus of the quasi-isotropic laminate E_{qi} = 55190 MPa
- Estimated ν_{23} truncated

PLAIN STRENGTHS

Ultimate stresses of the ply .

Property	Mean Values (MPa)				B-basis	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	70ºC/W	90ºC/W	K_B	-55C/Dry	70ºC/W	90ºC/W	-55C/Dry	RT/Dry	70ºC/W	90ºC/W
F_{1t}	-	2500	2500	2250	0.85	-	1.0	0.9	-	2100	2100	1900
F_{1c}	-	1500	900	750	0.80	-	0.6	0.5	-	1200	720	600
F_{2t}	-	45	27	22	0.80	-	0.6	0.5	-	36	22	18
F_{2c}	-	200	120	100	0.80	-	0.6	0.5	-	160	96	80
F_{12}	-	90	60	50	0.80	-	0.67	0.55	-	72	48	40

Design values for checking the laminate strength using the *Maximum Truncated Strain Criterion*.

Property	Mean Values (µε)				B-basis	E-KDF			B-basis Values (µε)			
	-55C/Dry	RT/Dry	70ºC/W	90ºC/W	K_B	-55C/Dry	70ºC/W	90ºC/W	-55C/Dry	RT/Dry	70ºC/W	90ºC/W
ε_{tall}	-	15000	15000	13500	0.85	-	1.0	0.9	-	12700	12700	11500
ε_{call}	-	10000	7000	6000	0.80	-	0.7	0.6	-	8000	5600	4800

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis	E-KDF			B-basis Values			
	-55C/Dry	RT/Dry	70ºC/W	90ºC/W	K_B	-55C/Dry	70ºC/W	90ºC/W	-55C/Dry	RT/Dry	70ºC/W	90ºC/W
F_{13} (MPa)	-	75	50	45	0.80	-	0.67	0.6	-	60	40	36
F_{3t} (MPa)	-	45	27	22	0.80	-	0.6	0.5	-	36	22	18
G_{IC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IIC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	70ºC/W	90ºC/W	K_B	-55C/Dry	70ºC/W	90ºC/W	-55C/Dry	RT/Dry	70ºC/W	90ºC/W
OHT_{QI}	-	550	550	550	0.85	-	1.0	1.0	-	470	470	470
OHC_{QI}	-	300	210	180	0.80	-	0.7	0.6	-	240	170	140
FHT_{QI}	-	550	550	550	0.85	-	1.0	1.0	-	470	470	470
FHC_{QI}	-	400	280	240	0.80	-	0.7	0.6	-	320	220	190
$F_{BR\text{-}QI}$	-	900	720	630	0.80	-	0.8	0.7	-	720	580	500
F_{PO}	-	-	-	-	-	-	-	-	-	-	-	-

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression (as well as approximation for dent depth to impact energy).

Property	B-basis Value for all conditions
TAI (µε)	6000

$$E/t^2 = A \cdot (1 - e^{-B \cdot d0/t})$$

Property	Design Value
A (J/mm^2)	5.0
B	5.0

Approximation model type: SMM chapter 1421: CDTM-T07

$$CAI = K_B \cdot EKDF \cdot CAI_{\infty\text{-}QI} \cdot (3 / K_{TOH})^C \cdot [\,1 + A / e^{B \cdot (E/t\hat{}2)}\,]$$

Property	Mean Value RT/Dry	B-basis K_B	*E-KDF* 70ºC/W	*E-KDF* 90ºC/W	*E-KDF* 120ºC/W
$CAI_{\infty\text{-}QI}$ (με)	3500	0.80	0.67	0.55	-
A	1.0				
B (mm^2/J)	0.5				
C	1.0				

…where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- K_{TOH} is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as:

$$K_{TOH} = 1 + \{\, 2 \cdot [(E_x/E_y)^{0.5} - v_{xy}] + E_x/G_{xy} \,\}^{0.5}$$

(*) CAI shall be calculated not only for the 0º direction (X parallel to material 0º reference), but also for the rest of ply orientations, i.e. the 45º and 90º directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x, E_y, G_{xy} and v_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the alowable strains applicable to plies at 0º (CAI_0), at ±45º (CAI_{45}) and at 90º (CAI_{90}).